

BCP for Countably Generated Continuous $C_0(X)$ -Algebras

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when its pure state space, i.e. K , has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Let A be a continuous $C_0(X)$ -algebra over a locally compact Hausdorff space X . Write A_x for the fiber and $a(x) \in A_x$ for the image of $a \in A$. We say that A is countably generated as a $C_0(X)$ -algebra if there is a sequence $(a_n) \subset A$ such that the C^* -algebra generated by $C_0(X)A$ and the a_n 's is all of A .

Theorem 1. *Let A be a continuous $C_0(X)$ -algebra. Assume that $A_x \neq 0$ for every $x \in X$, and that A is countably generated as a $C_0(X)$ -algebra. Then the following are equivalent:*

1. A has BCP;
2. A has UBCP;
3. X has a countable π -basis.

Idea of the Proof

The negative implication is the continuous-field base obstruction: if BCP holds, the near-norm open sets of any countable family of centers force a countable π -basis of X .

For the positive implication, countable generation supplies the missing noncommutative analogue of scalar directions. Rational $*$ -polynomials in the generating sections are countable and dense in every fiber. A polynomial section d might have large norm away from the fiber where it is useful, so we replace it by the bounded section

$$\sigma_d(x) = \frac{d(x)}{\max(\|d(x)\|, 1)}.$$

This pointwise radial normalization is still localizable by compactly supported scalar cutoffs, and it preserves fiberwise approximation to unit vectors up to a harmless factor. Thus countable generation yields the countable fiber-net needed for the bump-center UBCP construction.

Proof

We use two ingredients. The first is the base obstruction, proved in the companion packet [5].

Lemma 1 (base obstruction). *Let A be a continuous $C_0(X)$ -algebra with $A_x \neq 0$ for every $x \in X$. If A has BCP, then X has a countable π -basis.*

The second ingredient packages the countable local directions used in the previous fiber-net packet [6].

Definition 1. *Let $0 < \delta < 1$. A countable localizable δ -fiber-net for A is a countable family of bounded sections $\sigma_m(x) \in A_x$ such that:*

1. $\sup_x \|\sigma_m(x)\| \leq 1$;

2. $h\sigma_m \in A$ for every $h \in C_c(X)$;

3. for every $a \in A$, the map

$$(x, s) \mapsto \|a(x) - s\sigma_m(x)\|$$

$$\text{is continuous on } X \times [0, 4];$$

4. for every $x \in X$ and every $u \in S_{A_x}$, some m satisfies $\|\sigma_m(x) - u\| < \delta$.

Lemma 2 (fiber-net criterion). *Let A be a continuous $C_0(X)$ -algebra with all fibers nonzero. If A has a countable localizable $1/64$ -fiber-net and X has a countable π -basis, then A has UBCP.*

Proof. Let (V_j) be a countable π -basis. Choose $h_j \in C_c(X)$ with $0 \leq h_j \leq 1$, $\|h_j\|_\infty = 1$, and $\text{supp } h_j \subset V_j$. If (σ_m) is the fiber-net, form centers

$$c_{j,m} = 4h_j\sigma_m$$

and retain only those with $\|c_{j,m}\| > 15/4$. Use fixed radius $7/2$. These balls are uniformly off the origin because $15/4 - 7/2 = 1/4$.

Let $a \in S_A$. Pick x_0 with $\alpha = \|a(x_0)\| > 3/4$, set $u = a(x_0)/\alpha$, and choose m with $\|\sigma_m(x_0) - u\| < 1/64$. Convexity of

$$s \mapsto \|\alpha u - 4s\sigma_m(x_0)\|, \quad 0 \leq s \leq 1,$$

and the estimate

$$\|\alpha u - 4\sigma_m(x_0)\| \leq 4 - \alpha(1 - 1/64) < 7/2$$

give a strict $7/2$ bound at x_0 , uniformly for $s \in [0, 1]$. By the joint continuity in the fiber-net definition, this bound holds on a neighborhood U of x_0 ; after shrinking U , also $\|\sigma_m(x)\| > 1 - 2/64$ on U . Choose j with $V_j \subset U$.

For $x \in \text{supp } h_j$, apply the uniform estimate with $s = h_j(x)$; outside $\text{supp } h_j$, the center vanishes and $\|a(x)\| \leq 1$. Hence $\|a - c_{j,m}\| < 7/2$. Since h_j attains 1 inside U ,

$$\|c_{j,m}\| \geq 4(1 - 2/64) = 31/8 > 15/4,$$

so this center was retained. Thus the fixed-radius balls cover S_A , and they avoid $B(0, 1/4)$. This is UBCP. $\square$

It remains to show that countable generation gives a countable localizable fiber-net.

Lemma 3. *Let A be a countably generated continuous $C_0(X)$ -algebra. Then A has a countable localizable $1/64$ -fiber-net.*

Proof. Let (a_n) be a countable generating family. Let D be the countable set of all finite $(\mathbb{Q} + i\mathbb{Q})$ -linear combinations of nonempty words in finitely many variables a_n and a_n^* . For each x , the set

$$D(x) = \{d(x) : d \in D\}$$

is dense in A_x : the fiber A_x is generated by the images $a_n(x)$, and scalar rational $*$ -polynomials are dense in the generated fiber C^* -algebra.

For $d \in D$, define a scalar function

$$g_d(x) = \frac{1}{\max(\|d(x)\|, 1)}$$

and a bounded section

$$\sigma_d(x) = g_d(x)d(x).$$

Since A is continuous, $x \mapsto \|d(x)\|$ is continuous; hence g_d is continuous and bounded. Clearly $\|\sigma_d(x)\| \leq 1$ for all x .

If $h \in C_c(X)$, then $hg_d \in C_c(X)$, so

$$h\sigma_d = (hg_d)d \in A.$$

Thus σ_d is localizable.

We next verify the continuity condition. Fix $a \in A$, $d \in D$, and $(x_0, s_0) \in X \times [0, 4]$. Choose a relatively compact neighborhood U of x_0 and a function $\chi \in C_c(X)$ with $\chi = 1$ on U . On $U \times [0, 4]$,

$$a(x) - s\sigma_d(x) = a(x) - s(\chi g_d)(x)d(x).$$

For each fixed s , the section $a - s(\chi g_d)d$ belongs to A , and the norm function is continuous in x . Also, on U ,

$$\|a(x) - s\sigma_d(x)\| - \|a(x) - t\sigma_d(x)\| \leq |s - t| \|\sigma_d(x)\| \leq |s - t|.$$

Thus, at (x_0, s_0) , first keep $s = s_0$ and use continuity in x , then use this uniform Lipschitz estimate in the parameter s . Hence $(x, s) \mapsto \|a(x) - s\sigma_d(x)\|$ is continuous.

Finally, let $x \in X$ and $u \in S_{A_x}$. Choose $d \in D$ with $\|d(x) - u\| < 1/128$. If $\|d(x)\| \leq 1$, then

$$\|\sigma_d(x) - u\| = \|d(x) - u\| < 1/128.$$

If $\|d(x)\| > 1$, then

$$\begin{aligned} \|\sigma_d(x) - u\| &\leq \left\| \frac{d(x)}{\|d(x)\|} - d(x) \right\| + \|d(x) - u\| \\ &= |\|d(x)\| - 1| + \|d(x) - u\| \leq 2\|d(x) - u\| < 1/64. \end{aligned}$$

Thus the countable family $(\sigma_d)_{d \in D}$ is a localizable $1/64$ -fiber-net. $\square$

Proof of Theorem 1. UBCP implies BCP by definition. Lemma 1 gives $\text{BCP} \Rightarrow$ countable π -basis of X . Conversely, if X has a countable π -basis, Lemma 3 supplies a countable localizable $1/64$ -fiber-net, and Lemma 2 gives UBCP. $\square$

Scope and Limitations

This theorem is a substantial partial answer to the source question for continuous $C_0(X)$ -algebras, but it is not a full arbitrary C^* -algebra classification. The hypotheses of a continuous central base and countable generation over that base are used essentially. The proof does not cover merely upper-semicontinuous fields or simple algebras with no useful central $C_0(X)$ -structure.

Novelty and Search Bounds

The run indexes were searched for arXiv:2311.14257, C^* -algebras, BCP, UBCP, $C_0(X)$ -algebras, continuous fields, countable generation, and π -bases. Web searches on June 17, 2026 for countably generated $C_0(X)$ -algebra BCP formulations found no exact existing version of this criterion. The surrounding literature includes the source paper [1], the vector-valued $C_0(K, X)$ theorem [2], the Calkin-algebra obstruction [3], and the operator-space approximation criteria [4].

Human Review Focus

The key check is Lemma 3. In particular, verify that rational $*$ -polynomials in the generating sections are fiberwise dense, and that the pointwise normalized sections $\sigma_d = d/\max(\|d(\cdot)\|, 1)$ are localizable and have the stated joint continuity property.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Minzeng Liu, Rui Liu, Jimeng Lu, and Bentuo Zheng, *Ball covering property from commutative function spaces to non-commutative spaces of operators*, arXiv:2111.04921, 2021.
- [3] Qiyao Bao, Rui Liu, and Jie Shen, *The ball-covering property of non-commutative spaces of operators on Banach spaces*, arXiv:2412.07137, 2024.

- [4] Qiyao Bao, Rui Liu, and Jie Shen, *Approximation properties and quantitative estimation for uniform ball-covering property of operator spaces*, arXiv:2507.02261, 2025.
- [5] Run `fa_banach_001`, lane 19, *A Base-Space Obstruction for Continuous $C_0(X)$ -Algebras with BCP*, solution packet `2311.14257_c0x_algebra_base_pi_obstruction`, 2026.
- [6] Run `fa_banach_001`, lane 19, *A Fiber-Net Criterion for BCP in Continuous $C_0(X)$ -Algebras*, solution packet `2311.14257_continuous_field_fibernet_bcp`, 2026.